

The empirical recurrence for OEIS A317867, recovered from its tail

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Abstract

OEIS A317867 records “Empirical recurrence of order 83” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 105$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{83} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-83 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317867 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 2, 3 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 2, 6, 26, 88, 301, 1089, 4340, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Aug 09 2018 09:53:36, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 105$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 105$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 83: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 83.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 210$ exact terms from $n = 4$ on, it returns order 83 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	1	c_{22}	−31254	c_{43}	−419883	c_{64}	−18807
c_2	8	c_{23}	22759	c_{44}	−99235	c_{65}	−3550
c_3	7	c_{24}	−821	c_{45}	−40211	c_{66}	−3219
c_4	19	c_{25}	50827	c_{46}	317166	c_{67}	4966
c_5	24	c_{26}	−49852	c_{47}	−148785	c_{68}	908
c_6	−59	c_{27}	−108942	c_{48}	−254250	c_{69}	721
c_7	−150	c_{28}	−152106	c_{49}	38094	c_{70}	−2705
c_8	−294	c_{29}	−90338	c_{50}	126694	c_{71}	−1078
c_9	−269	c_{30}	12428	c_{51}	133012	c_{72}	−475
c_{10}	−560	c_{31}	−28825	c_{52}	134004	c_{73}	865
c_{11}	967	c_{32}	92629	c_{53}	118188	c_{74}	315
c_{12}	−249	c_{33}	165350	c_{54}	−123877	c_{75}	495
c_{13}	1797	c_{34}	370529	c_{55}	−43657	c_{76}	128
c_{14}	−2833	c_{35}	−49641	c_{56}	−71450	c_{77}	176
c_{15}	2480	c_{36}	−20280	c_{57}	5086	c_{78}	−59
c_{16}	−8549	c_{37}	170382	c_{58}	80708	c_{79}	−78
c_{17}	12499	c_{38}	279072	c_{59}	65968	c_{80}	−57
c_{18}	1630	c_{39}	320020	c_{60}	13587	c_{81}	−3
c_{19}	27748	c_{40}	−274473	c_{61}	10900	c_{82}	8
c_{20}	−14906	c_{41}	−249198	c_{62}	−23178	c_{83}	4
c_{21}	15337	c_{42}	−330108	c_{63}	−14721		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A317867 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{83} - c_1x^{83-1} - \dots - c_{83}$. Its constant term is $-c_{83} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 83 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 83, so $\deg q = 83$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 83, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 83, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-83 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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